

Final Project Scenario 1: Understanding Diabetes Progression

1. Understand and prepare the data

The dataset tracks 442 patients, with each row representing an individual's medical profile. The primary outcome is progression (originally Y), a continuous numerical measure of diabetes progression one year after baseline. The data includes 10 predictors: age, sex, BMI, blood pressure (BP), and six numerical serum measurements (S1-S6). Sex is encoded numerically (1 and 2), but it is a categorical variable and was converted to a factor for appropriate analysis. The dataset is exceptionally clean, with zero missing values (NAs) and zero duplicated rows. Analyzing health data requires caution: patients' privacy must be protected, and the statistical relationships identified in the data are observational, meaning they do not necessarily reflect direct medical causation.

2. Explore the data

The disease progression variable has a mean of 152.1 and a median of 140.5, with values ranging widely from a minimum of 25 to a maximum of 346. BMI averages 26.4 with a standard deviation of 4.42 (ranging from 18.0 to 42.2). The patient ages range from 19 to 79, with an average age of 48.5 years. Visualizing the distributions reveals that the progression measure is right-skewed, meaning a smaller subset of patients experienced much more severe disease progression. BMI roughly follows a normal distribution but includes high-end outliers past the 35 mark, which represents severely obese patients.

3. Investigate relationships

A correlation matrix of the numerical variables reveals that BMI ($r = 0.59$), serum measurement S5 ($r = 0.57$), and blood pressure ($r = 0.44$) are the most strongly associated variables with positive disease progression. Conversely, serum measurement S3 is negatively correlated with progression ($r = -0.39$). Because BMI, S5, and BP exhibit the strongest linear relationships with the outcome without being perfectly correlated with one another, they are the most promising candidates for multiple regression modelling.

4. Build and interpret regression models

A simple linear regression predicting progression using only BMI yields the fitted equation: $Progression = -117.77 + 10.23(BMI)$. This means that for every 1-unit increase in BMI, a patient's disease progression measure increases by 10.23 points on average. The R^2 is 0.344, meaning BMI alone explains 34.4% of the variance in disease progression. The intercept (-117.77) lacks real-world interpretation, as a BMI of zero is impossible.

A multiple regression model predicting progression based on BMI, BP, and S5 yields an R^2 of 0.480, meaning this combined model accounts for roughly 48% of the variance in progression. Holding blood pressure and S5 constant, a 1-unit increase in BMI is associated with a 6.50-point increase in progression. The multiple model offers a notably stronger fit than the simple model, capturing a wider array of physiological factors driving the disease.

5. Evaluate the analysis

The multiple model's diagnostic Residuals vs. Fitted plot displays relatively random scatter, though the variance slightly fans out at higher fitted values, suggesting mild heteroscedasticity. Variance Inflation Factor (VIF) analysis shows no concerning multicollinearity among the predictors: BMI, BP, and S5 all have VIF values around 1.3 (well below the common warning threshold of 5). The statistical uncertainty is very low for our key coefficients: the p-values for BMI, BP, and S5 are all less than 0.001, confirming they are statistically significant. However, statistical significance is not the same as practical importance; while these predictors are scientifically valid markers, an R^2 of 0.48 indicates that over half of the variation in diabetes progression is still driven by unmeasured factors (like diet, genetics, or medication adherence).

6. Draw appropriate conclusions

This dataset is observational, not experimental. We cannot interpret the regression coefficients causally; we can only state that higher BMI, BP, and S5 are *associated* with worse progression, not that they directly *cause* it. There are almost certainly confounding variables missing from this dataset, such as exact diet, lifestyle habits, or pre-existing comorbidities. Furthermore, we do not know the demographic or geographic makeup of these 442 patients, which limits how

confidently we can generalize these findings to the broader global population. We can safely conclude that BMI and S5 are strong warning indicators for diabetes progression risk, but we cannot conclude that modifying these factors will guarantee a slower disease progression.